

Enucleation after Plaque Radiotherapy for Posterior Uveal Melanoma

Histopathologic Findings

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Abstract: A review of 1019 patients with posterior uveal melanomas, who were treated with episcleral plaque radiotherapy between April 1976 and December 1987, showed that 59 (6%) of these patients have thus far required enucleation of the affected eye. The most common clinical reasons for enucleation were tumor regrowth (51%) and neovascular glaucoma (31%). These irradiated melanomas were of mixed cell type in 40 cases (68%), spindle in 10 cases (17%), totally necrotic in 7 cases (12%), and epithelioid in 2 cases (3%). Some degree of tumor necrosis was seen in all cases. Overall, the median number of mitoses per 40 high-power fields in these irradiated tumors was 0. Most (64%) of the tumors had no identifiable mitosis in 40 high-power fields. A three-way simultaneous comparison of the following groups was made: irradiated eyes enucleated for tumor regrowth, irradiated eyes enucleated for neovascular glaucoma, and a matched group of nonirradiated posterior uveal melanomas. This comparison showed that the largest tumor dimension was significantly greater in the irradiated eyes enucleated for tumor regrowth than in the other two groups. Mitotic activity was significantly higher in the irradiated tumor regrowth and nonirradiated groups than in the irradiated neovascular group. There was no significant difference among the three groups in an analysis of height. The results suggested that mitotic activity may exist after plaque radiotherapy, especially when there is clinical evidence of tumor regrowth.

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Episcleral plaque radiotherapy has proven to be an effective method of controlling the growth of melanomas of the ciliary body and choroid.¹⁻¹⁰ In the most cases, the tumor responds to this treatment by gradually decreasing in size over a period of months to years.^{8,10} In most cases the tumor shows excellent regression and the affected eye is retained, often with good vision. Such eyes are not enucleated, and consequently little histopathologic material is available. Unfortunately, in some cases regrowth of the tumor or severe complications of the radiotherapy may necessitate enucleation of the affected eye.¹¹⁻¹⁶

There is little information in the literature on the histopathologic findings of plaque-irradiated uveal mela-